

Stance Lens: Evaluating Large Language Models for Arabic Stance Detection on Women's Sports in Saudi Arabia

Areej Alsaiani
Lamar Allihyani
Shahad Alsaiani
Noura Alhuthali (Leader)

**Data Science Department
College of Computing
Umm Al-Qura University**

2 June 2026

This project report is submitted to the Department of Data Science at Umm Al-Qura University in partial fulfilment of the requirements for the degree of Bachelor of Science in Data Science.

Students:

Areej Alsaiari 444002403

Lamar Allihyani 444007229

Shahad Alsaiari 444005703

Noura Alhuthali 444001743

Supervisor(s):

supervisor's name: Omaina Fallatah

email: oafallatah@uqu.edu.sa

Intellectual Property Right Declaration

This is to declare that the work under the supervision of Dr Omaina Fallatah having title " Stance Lens: Evaluating Large Language Models for Arabic Stance Detection on Women's Sports in Saudi Arabia" carried out in partial fulfilment of the requirements of Bachelor of Science in 2025/2026, is the sole property of the Umm Al Qura University and the respective supervisor and is protected under the intellectual property right laws and conventions. It can only be considered/used for purposes like extension for further enhancement product, development, adoption for commercial/organizational usage, etc., with the permission of the University and respective supervisor. This above statement applies to all students and faculty members.

Date:2 – 6 – 2026.....

Author(s):

Name: Noura Alhuthali

Name: Areej Alsaiari

Name: Lamar Allihyani

Name: Shahad Alsaiari

Signature: -----

Signature: -----

Signature: -----

Signature: -----

Supervisor(s):

Name: Omaina Fallatah

Anti-Plagiarism Declaration

This is to declare that the above publication produced under the supervision of Dr. Omaima Fallatah having title " Stance Lens: Evaluating Large Language Models for Arabic Stance Detection on Women's Sports in Saudi Arabia" is the sole contribution of the author(s) and no part hereof has been reproduced illegally (cut and paste) which can be considered as Plagiarism. All referenced parts have been used to argue the idea and have been cited properly. We will be responsible and liable for any consequence if violation of this declaration is proven.

Date:2 – 6 – 2026.....

Author(s):

Name: Noura Alhuthali

Name: Areej Alsaiari

Name: Lamar Allihyani

Name: Shahad Alsaiari

Signature: -----

Signature: -----

Signature: -----

Signature: -----

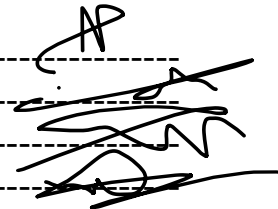
The image shows four handwritten signatures in black ink, each written over a horizontal dashed line. The signatures are stylized and cursive. The first signature is the simplest, while the others are more complex and overlapping.

Table of Contents

Intellectual Property Right Declaration	3
Anti-Plagiarism Declaration	4
Table of Contents	5
List of Tables	7
List of Figures	8
Abstract	9
Chapter 1 Introduction	10
1.1 Project Motivation	10
1.2 Problem Statement	10
1.3 Project Contribution	10
1.4 Structure of the Project	11
Chapter 2 Related Work	12
2.1 Stance Detection in NLP	12
2.2 Traditional Machine Learning Approaches	12
2.3 Transformer-Based Deep Learning Models	13
2.4 Large Language Model (LLM) Adaptation	14
2.5 Challenges in Arabic Stance Detection	14
2.6 Summary of Related Studies on Arabic Stance Detection	16
Chapter 3 Methodology	17
3.1 Method Overview	17
3.1.1 Data Collection	17
3.1.2 Data Annotation	18
3.1.3 Data Preprocessing	19
3.1.4 Modelling	21
Chapter 4 Results and Discussion	23
4.1 Evaluation Metrics	23
4.2 Results	24
4.3 Error analysis	27
4.3.1 AraBERT Errors	29
4.3.2 ALLaM Errors	30
4.3.3 LLaMA Errors	31
4.3.4 LoRA-ALLaM Errors	32
4.4 Limitations	32
4.4.1 Data Scarcity and Arabic Language Representation	32
4.4.2 Annotation Complexity and Subjectivity	33

4.4.3 Prompt Design	33
Chapter 5 Conclusion and Future Work	34
References	35
Appendixes	36

List of Tables

Table 1 Summary of Related Studies on Arabic Stance Detection	16
Table 2 Summary of Dataset Statistics After Annotation.....	19
Table 3 Performance Comparison of Evaluated Models for Arabic Stance Detection.....	24
Table 4 Detailed Per-Class Performance of All Models Using Precision, Recall, and F1-Score	26

List of Figures

Figure 1: Overall Research Framework for Stance Detection17

Figure 2 Confusion Matrices of AraBERT and ALLaM and LLaMA-3shot25

Figure 3 Distribution of Classification Errors Across Models27

Figure 4 Classes Contributing Most to Misclassification28

Figure 5 Error Analysis per Class Across Different Models28

Figure 6 Sample Misclassified Tweets by AraBERT29

Figure 7 Sample Misclassified Tweets by ALLaM30

Figure 8 Sample Misclassified Tweets by LLaMA 3.131

Figure 9 Sample Misclassified Tweets by LoRA-ALLaM.....32

Abstract

In recent years, women's sports in Saudi Arabia have gained significant visibility and public attention, leading to increased discussions on social media platforms, particularly X (formerly Twitter) and YouTube. These discussions reflect a wide range of viewpoints, including supportive opinions that encourage women's participation in sports, opposing views that criticize or reject the idea, and neutral expressions that do not present a clear stance. This project aims to develop an Arabic stance detection system capable of automatically classifying social media content related to women's sports into three categories: Supportive, Opposing, and Neutral. The dataset will be collected from both X and YouTube using relevant keywords and hashtags, then cleaned, preprocessed, and manually annotated to ensure high-quality labels. To evaluate the effectiveness of different computational approaches, the study compares deep learning models, and a large language model (LLM) using prompt-based classification. Model performance will be assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score. Overall, this work provides a comprehensive comparison of deep learning, and LLM-based methods for Arabic stance detection, while offering insights into Arab public discourse on women's sports in Saudi Arabia using modern NLP techniques.

Keywords: Natural language processing · Arabic · large language models (LLMs) Stance detection ·

Chapter 1 Introduction

1.1 Project Motivation

Saudi Arabia has recently witnessed significant progress in empowering women, especially in sports, as part of Vision 2030. While these changes have opened new opportunities, the public's attitude toward women's sports remains a topic of social debate. This project aims to evaluate the capability of large language models (LLMs) to understand stance in Arabic text, particularly in discussions about Saudi women's sports. By analysing Arabic tweets using advanced Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques, the study seeks to assess how effectively LLMs can detect whether a stance is supportive, opposing, or neutral. The findings are expected to provide insights into both public opinion and the performance of modern AI models in processing Arabic social discourse.

1.2 Problem Statement

Despite the growing attention toward women's participation in sports in Saudi Arabia, there is still limited understanding of how the public perceives this change. Most existing studies focus on policy or participation rates but overlook real social opinions expressed on digital platforms.

The core problem addressed in this project is the lack of an automated and accurate method to analyse Arabic social media content regarding women's sports. Without such analysis, it becomes challenging to objectively assess public stance and identify patterns of support, opposition, or neutrality. This project aims to fill this gap by developing an LLM model that applies Natural Language Processing (NLP) techniques to classify Arabic tweets into stance categories. The outcomes are expected to contribute to the academic domain by introducing a data-driven approach to understanding social attitudes, while also supporting decision-makers and organizations in promoting women's empowerment in the Saudi sports sector.

Research Questions:

1. How effectively can stance detection models classify Arabic tweets related to women's sports in Saudi Arabia?
2. What are the differences in performance between deep learning models, and large language models (LLMs) for stance detection.

1.3 Project Contribution

Project contributes to both the academic and practical understanding of social attitudes toward women's sports in Saudi Arabia by developing an LLM model that applies Natural Language Processing (NLP) techniques to analyze Arabic social media content, Addressing the identified gap of lacking automated and accurate tools for Arabic stance analysis, this study

introduces an LLM-based framework capable of classifying Arabic tweets into *supportive*, *opposing*, or *neutral* categories.

From a research perspective, the project advances the field of Arabic NLP by developing a tailored stance detection model for Arabic social discourse, focusing on culturally and linguistically rich contexts. It also highlights the ability of Large Language Models (LLMs) to understand subtle and complex social opinions in Arabic text.

From a practical perspective, the project focuses on applying and fine-tuning a Large Language Model (LLM) specifically for Arabic stance detection in the context of women's sports. This includes improving the model's performance in analyzing Arabic content and adapting it to the Saudi social and cultural context. The developed model is expected to serve as an effective AI-driven tool for understanding public opinion and guiding data-informed strategies. The outcomes of this project align with the goals of Saudi Vision 2030, supporting women's empowerment and participation in sports and providing valuable insights for policymakers and organizations to foster awareness and positive engagement in women's sports.

1.4 Structure of the Project

This project explores the analysis of public stance toward women's sports in Saudi Arabia using Large Language Models (LLMs). The research reviews previous studies on stance detection and Arabic NLP, highlighting recent advancements in LLMs and their applications in Arabic text analysis. It outlines the methodology for collecting data from Arabic Twitter posts and YouTube videos, along with preprocessing techniques, and the development of an LLM-based stance detection model. The results showcase the model's ability to classify stances as supportive, opposing, or neutral. The project highlights the effectiveness of LLMs in analyzing social media content and suggests future improvements, including fine-tuning the model and expanding datasets.

Chapter 2 Related Work

2.1 Stance Detection in NLP

Stance detection is a Natural Language Processing (NLP) task that identifies whether the author's viewpoint in a text is in favor of, against, or neutral toward a specific topic. Unlike sentiment analysis, which measures emotional tone, stance detection focuses on opinions directed toward a particular subject. This approach is especially useful for analyzing social discussions expressed through short, informal texts such as tweets, where language is often dialectal and context dependent [1].

Stance detection plays a crucial role within the broader field of Natural Language Processing (NLP) as it enables machines to interpret opinions and perspectives embedded in textual content, rather than merely detecting sentiment polarity. Its importance lies in enhancing computational understanding of argumentation, persuasion, and ideology across digital communication platform [2]. Over the past few years, stance detection has been widely applied in various domains, including public health discussions during the COVID-19 pandemic, where it was used to identify support or opposition to vaccination campaigns; political debates, to track ideological alignment and polarization; and social movements, such as gender equality and climate activism, to analyze public attitudes on sensitive issues [3],[4]. These applications demonstrate how stance detection contributes to more context-aware, socially informed NLP systems that can interpret complex, opinion-rich online discourse.

2.2 Traditional Machine Learning Approaches

Initially, Alkhraiji and Azmi [1]. study introduced a stance detection framework for Arabic tweets discussing Hezbollah's activities in Syria, aiming to identify content and classify stance. The study addressed Arabic linguistic challenges using preprocessing with the ISRI stemmer and TF-IDF vectorization. The authors employed traditional machine learning models—Multinomial Naive Bayes, SVM, and weighted KNN—while acknowledging that deep learning models often achieve higher accuracy. They intentionally focused on simpler models to ensure interpretability, computational efficiency, and robustness with limited training data, priorities that are crucial in counterterrorism contexts where transparency and explainability are essential. Performance was compared on both imbalanced data and a balanced version using SMOTE, which improved results to F1-score of 78.6%. While the study provides a transparent and effective baseline for Arabic stance detection, its reliance on small datasets and non-deep-learning models limits scalability and potential performance. Overall, the research offers valuable insights into future work, particularly for building more advanced models or using larger datasets.

2.3 Transformer-Based Deep Learning Models

Advancing transformers, Alkathlan et al. [4]. developed ArabicStanceX, a large-scale Arabic stance detection dataset sourced from X (formerly Twitter), consisting of 14,477 tweets across 17 topics. Using MARBERTv2 with multi-topic fine-tuning strategies, the study achieved an F1-score of 0.74 for favor/against stances. However, the model faced several challenges, including difficulty detecting neutral stances, class imbalance across topics, and limited generalization to unseen or sensitive subjects such as religion or social issues. The authors also noted that GPT-4-generated topic descriptions were less effective than manually written ones. Overall, the dataset represents a significant step toward advancing Arabic NLP and stance detection research, particularly within the Saudi social media context.

In multi-dialectal settings, Charfi et al. [5]. introduced *MARASTA*, a multi-dialectal Arabic stance detection corpus that includes 4,657 sentences collected from various online platforms across four major Arab regions — the Gulf, Egypt, the Levant, and the Maghreb — written in both Modern Standard Arabic and local dialects. Each instance was manually annotated into three stance categories: *Pro*, *Against*, and *Neutral*. The authors tested several models, and AraBERT achieved the best performance with an F1-score of 0.82, outperforming classical approaches like SVM and Logistic Regression. The study revealed that dialectal variation strongly affects model performance, with the highest accuracy in Levantine Arabic (0.89) and the lowest in Gulf Arabic (0.75). Main challenges included difficulty in detecting neutral stances, class imbalance across dialects, and limited contextual understanding since the dataset is sentence-based. Despite these limitations, *MARASTA* represents an important advancement in Arabic NLP, providing a balanced and diverse resource that supports research on cross-dialect stance detection and highlights the importance of considering linguistic diversity in Arabic social media analysis.

Within shared task frameworks, Alturayef et al. [6]. Introduced StanceEval 2024, the first large-scale Arabic stance detection shared task, designed to establish a common evaluation framework for Arabic NLP systems. The organizers developed the MAWQIF dataset, containing over 4,000 annotated tweets related to women's empowerment, COVID-19 vaccination, and digital transformation. Each tweet was labeled for stance (favor, against, or none), sentiment, and sarcasm, encouraging multi-dimensional model training. Twenty-eight teams participated in the competition, with the best-performing system—based on AraBERT—achieving an F1-score of 0.84 through multi-task learning and data augmentation. The study demonstrated the strength of transformer-based Arabic models and emphasized the benefits of jointly modeling sentiment and stance. It also provided the first standardized benchmark for Arabic stance detection. Critically, while the shared task unified evaluation protocols, its dataset was limited to three topics and one platform (Twitter), constraining cross-domain generalization.

2.4 Large Language Model (LLM) Adaptation

Models (LLMs) are advanced neural networks trained on massive text corpora to understand and generate human-like language. Examples include GPT-4, Llama, and BERT, which have shown outstanding performance across a wide range of Natural Language Processing (NLP) tasks such as text classification, sentiment analysis, translation, and question answering. Their ability to generalize across languages and domains has revolutionized the NLP field, making them a foundation for modern language understanding and generation systems.

At the LLM stage, Ibrahim [7]. explored the potential of large language models (LLMs) for Arabic stance detection by fine-tuning LLaMA-3 (8B parameters) using LoRA (Low-Rank Adaptation) on the MAWQIF dataset. Despite LLaMA-3 not being originally trained for Arabic, parameter-efficient fine-tuning and 4-bit quantization allowed the model to learn Arabic stance cues effectively while minimizing computational cost. The model achieved a macro F1-score of 0.76, with higher accuracy on topics like women's empowerment and lower performance on digital transformation, suggesting topic sensitivity in stance learning. This work demonstrated that LLMs can adapt to Arabic with limited data and resources, highlighting Lora's practicality for academic and low-budget environments. However, the model's performance remained slightly below that of Arabic-specific transformers such as AraBERT, indicating that linguistic specialization and domain-specific data are still essential for optimal Arabic stance detection.

2.5 Challenges in Arabic Stance Detection

Implicitness of Targets: A key challenge in stance detection is the implicitness of targets, where the entity or issue toward which the stance is expressed is not explicitly mentioned in the text. Systems must infer stance from indirect references, rhetorical devices, or context-specific cues, which require advanced contextual and semantic understanding. [1].

Linguistic Diversity in Arabic: Arabic stance detection faces challenges due to the significant variation in dialects and vocabulary, along with the complex linguistic structure and rich contextual meaning, which can make it difficult to accurately determine stances across different Arabic texts. [2].

Lack of Available Datasets: There is a shortage of comprehensive datasets for Arabic stance detection, which limits the effectiveness of AI models in this area. The available datasets are typically small and not diverse enough to train models efficiently. [4].

Multi-task Learning Integration: Integrating multi-task learning, such as sentiment and sarcasm detection alongside stance detection, presents a challenge. It requires balancing multiple tasks and assigning appropriate weights to each. [4].

Annotation Disagreement: Human annotators frequently disagree on stance labels, especially for vague or sarcastic sentences. Resolving these conflicts requires additional annotation of rounds or discussion meetings. [5].

Handling Neutral Stances (None): Accurately classifying neutral stances remains a significant challenge, as many tweets do not explicitly show support or opposition to a topic, making it harder to categorize them correctly. [6].

Sarcasm Issue: Sarcasm is one of the biggest challenges in stance detection, especially with sensitive topics like the COVID-19 vaccine. Sarcasm can mislead sentiment analysis and stance classification, making it difficult for models to interpret accurately. [6].

large language model: While previous studies have addressed stance detection challenges in various Arabic dialects, none have examined them through the lens of Large Language Models (LLMs). This work is the first to explore how LLMs perform on Saudi dialect data, identifying the unique challenges and limitations of applying such models to stance detection in this linguistic context. [7].

A key challenge in applying LLMs to multi-class stance detection is label bias. Aiyappa et al. (2024) demonstrated that LLMs performing zero-shot stance detection exhibit a positivity bias, systematically favoring labels with positive connotations while under-predicting neutral or opposing stances. This behaviour is particularly problematic in three-class settings where the neutral class is already underrepresented, leading to near-zero recall for that class. [8]

2.6 Summary of Related Studies on Arabic Stance Detection

Study	Year	Methodology / Model	Dataset & Size	Main Findings (F1-score)	Key Contribution	Main Limitations	Relevance to Current Project
Alkhraiji & Azmi	2024	Traditional ML (SVM, NB, KNN) with TF-IDF & ISRI stemmer	~7,000 Arabic tweets about Hezbollah in Syria	78.6% (after SMOTE)	Interpretable ML baseline for Arabic stance detection	Small dataset; lacks deep learning depth	Good baseline before applying deep learning on sports data
Alkhatlan et al. (ArabicStanceX)	2025	Transformer (MARBERTv2) with multi-topic fine-tuning	14,477 tweets across 17 Saudi topics	0.74 (favor/against)	Large-scale Saudi stance dataset using transformers	Struggles with neutral stance & topic generalization	Useful for Saudi Twitter data collection & stance categories
Charfi et al. (MARASTA)	2024	Transformer (AraBERT)+ Classical ML comparison	4,657 sentences (4 dialects: Gulf, Egypt, Levant, Maghreb)	0.82 (AraBERT)	First multi-dialectal Arabic stance corpus	Dialect imbalance; limited context	Adapting model to Saudi dialect tweets
Alturayef et al. (StanceEval 2024)	2024	Shared Task using Transformer (AraBERT, Multi-task Learning)	4,000 tweets (MAWQIF dataset)	0.84 (best team)	Unified benchmark for Arabic stance detection	Limited to 3 topics; Twitter only	Reference for evaluation metrics & multi-task stance analysis
Ibrahim	2024	LLM (LLaMA-3 8B) fine-tuned with LoRa	MAWQIF dataset	0.76 (macro F1)	LLM adaptation for Arabic via LoRa fine-tuning	Slightly weaker than Arabic-specific transformers	Shows how to fine-tune large models efficiently (LoRa)

Table 1 Summary of Related Studies on Arabic Stance Detection

Overall, several studies have explored stance detection in Arabic social media, none have specifically focused on women's sports within the Saudi context. Moreover, there is currently no publicly available dataset addressing this topic or capturing the linguistic diversity of Saudi dialects to fill this gap, our project manually collects and labels Arabic tweets about women's sports in Saudi Arabia, then applies Large Language Models (LLMs) to evaluate the ability of different LLM on the task of automatic stance detection. This provides a unique dataset and valuable insights into how AI handles Saudi dialects and social attitudes.

Chapter 3 Methodology

3.1 Method Overview

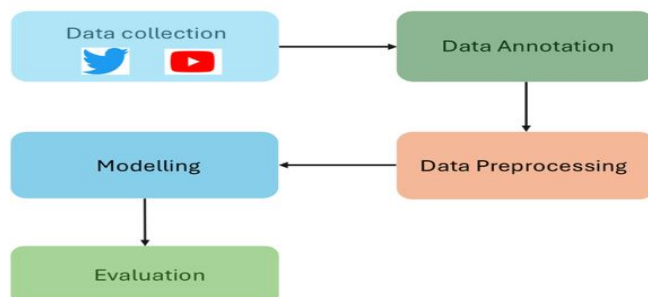


Figure 1: Overall Research Framework for Stance Detection

The project began with collecting data from Twitter and YouTube, focusing on posts and comments related to women's sports in Saudi Arabia. The data was then organized and initially cleaned to prepare it for annotation. Next, a manual labeling process was conducted in which three annotators independently classified each text as positive, negative, or neutral to ensure high accuracy and inter-annotator agreement. After annotation, the Data Preprocessing stage involved cleaning the content while intentionally preserving emojis, as they play an important role in detecting sarcasm and understanding the overall tone of the text. Unnecessary links, hashtags, and noise were removed as well. Following this, a Large Language Model (LLM) was utilized, and prompting techniques were applied to generate richer linguistic representations and enhance the text preparation. Based on these processed inputs, the system model was developed to understand deeper contexts rather than simple surface-level classification. Finally, the model was evaluated using metrics such as accuracy and F1-score to ensure the quality, effectiveness, and reliability of its performance within the proposed system.

3.1.1 Data Collection

Data were collected from Twitter (X) and YouTube to analyze Arab public opinions regarding women's participation in sports in Saudi Arabia. The aim of this study is not only to perform stance detection, but also to evaluate and enhance the performance of Large Language Models (LLMs) in understanding Arabic social context and sentiment.

YouTube comments were retrieved by extracting the comment sections directly from selected sports-related video URLs. This method provided longer and more conversational expressions, balancing the dataset with more diverse and detailed opinions compared to the short-style discourse typical of Twitter. For Twitter data, the Playwright tool was used to

collect publicly available Arabic tweets. This approach was selected because it allows efficient data collection without requiring paid API access, which was particularly practical during the period when Twitter API restrictions and pricing were high. Tweets were gathered using relevant Arabic keywords and hashtags such as:

#الرياضة النسائية , #المرأة_في_الرياضة , #دوري_السيدات , #المنتخب_النسائي_السعودي

After applying cleaning and filtering steps, approximately 1350 Arabic comments were retained. Duplicate and irrelevant content was removed to ensure dataset quality. The final dataset provides a foundation for training stance detection models and benchmarking Arabic Large Language Models (such as AraBERT and Llama) on real-world social discourse, thereby contributing both to understanding public attitudes and to advancing Arabic NLP capabilities.

3.1.2 Data Annotation

The dataset was manually annotated to determine the stance expressed toward women's participation in sports. Three annotators independently reviewed each text and labeled it as Positive (Pos), indicating supportive opinions; Negative (Neg), representing oppositional views; or Neutral (Net), for comments without a clear stance. This includes comments that were purely informational, unrelated to the main topic, focused on gameplay or general discussion, or written as questions without expressing an opinion. Manual annotation was chosen to ensure accurate interpretation of tone, cultural context, and figurative expressions common in Arabic social media. After completing the labeling process, inter-annotator agreement was calculated to evaluate the consistency among annotators and confirm the reliability of the final labeled dataset. The three annotators achieved a 77.15 % overall agreement, with a 22.85 % disagreement rate, indicating a *substantial* level of consistency. The value aligns with prior research on Arabic stance detection, where moderate to substantial agreement is commonly reported. Jaziri yan et al. [9] (2022) reported a Cohen's Kappa of approximately 0.66 in the *Resistance* corpus, classifying it as *substantial* according to the criteria of Landis and Koch (1977). Such results confirm that the manual annotation process in this study achieved a reliable level of agreement.

VALUE	METRIC
1348	Total number of comments
550	Positive comments (Pos)
610	Negative comments (Neg)
183	Neutral comments (Nat)
1040	Full agreement between annotators
77.15%	inter-annotator agreement percentage
308	comments with disagreement
22.85%	Disagreement percentage

Table 2 Summary of Dataset Statistics After Annotation

3.1.3 Data Preprocessing

After annotation, the content of the dataset is pre-processed to ensure data quality and suitability for model training. Non-informative elements, such as URLs, user account names, and hashtags, are being removed to preserve privacy and eliminate noise unrelated to stance meaning. Additionally, duplicate entries and irrelevant or off-topic comments are being filtered out to maintain a cleaner and more representative dataset. Unlike other text cleaning procedures, emojis are intentionally retained, as they play a significant role in conveying emotional tone and stance in Arabic social media discourse, particularly when textual expressions are short or indirect.

The final cleaned dataset is now ready for Natural Language Processing (NLP) tasks, where tokenization, embedding, and model training will be applied. This preprocessing step is essential to improve the model's ability to learn semantic patterns and accurately detect stance in real-world Arabic social contexts.

3.1.4 Experimental Setup

All experiments were conducted using **Google Collab** Environment **hosted runtime** with a **CUDA-enabled NVIDIA Tesla T4 GPU** and **High-RAM configuration**. The implementation was developed in Python using PyTorch and the Hugging Face ecosystem.

Main libraries used:

- PyTorch
- Hugging Face Transformers
- Hugging Face Datasets
- Hugging Face Hub (for accessing pretrained models)
- PEFT (Parameter-Efficient Fine-Tuning)
- Accelerate
- BitsAndBytes (4-bit quantization)
- Scikit-learn

A fixed random seed of **42** was used to ensure reproducibility.

for the models Settings in AraBERT Baseline (Frozen Embeddings) The Model was **aubmindlab/bert-base-arabertv02** and this is the following Procedure:

- AraBERT was used as a **frozen encoder**.
- Mean pooling was applied to obtain sentence embeddings.
- A **Logistic Regression** classifier was trained on top.

Key settings:

- Max sequence length: 128
- Batch size (embedding extraction): 16
- Logistic Regression: max_iter = 5000, class_weight = balanced, C = 0.2.

In Large Language Models (**Prompting and Inference Setup**) Two instruction-tuned LLMs were evaluated using constrained decoding:

- allam-ai/ALLaM-7B-Instruct-preview
- meta-llama/Llama-3.1-8B-Instruct

Pretrained models were accessed through the **Hugging Face Hub** and executed using the Transformers library. And Models were evaluated using zero-shot, one-shot, and three-shot

prompting. The task was formulated in Arabic, where the models were constrained to generate a single-token label from: "Pos": "إيجابي", "Neg": "سلبي", "Nat": "محايد".

To ensure deterministic classification, constrained decoding was applied by selecting the label with the highest next-token logit. And to enable inference on limited hardware, models were loaded using 4-bit quantization, float16 precision, and automatic device mapping.

To compare LLM fine-tuning against AraBERT, ALLaM was adapted for sequence classification using **LoRA**.

Model: **allam-ai/ALLaM-7B-Instruct-preview**

Quantization

- 4-bit loading using BitsAndBytes
- k-bit training preparation applied

LoRA Configuration

- Rank (r): 32
- LoRA alpha: 64
- LoRA dropout: 0.05
- Target modules: q_proj, k_proj, v_proj, o_proj

Training Hyperparameters

- Epochs: 5
- Batch size: 16
- Learning rate: 2e-4
- Warmup ratio: 0.1
- FP16 training enabled

Due to class imbalance, **class-weighted cross-entropy loss** was used during fine-tuning.

3.1.5 Modelling

In this phase of the project, we aim to test the capability of large language models (LLMs) like Llama and ALLaM in analysing the data accurately. We will compare the performance of these models with traditional deep learning models like BERT in the task of Arabic stance detection.

BERT - (Bidirectional Encoder Representations from Transformers) is a deep learning model for natural language processing. We will use BERT as a baseline, specifically AraBERT,

a version fine-tuned for Arabic text, to improve performance in tasks like Arabic stance detection.

LLM - (Large Language Models) are advanced models trained in vast amounts of text data, capable of understanding and generating human-like text. In our project, we will compare LLMs with the baseline model, BERT, to evaluate their performance, especially in handling Arabic texts, with a focus on Saudi Arabic. This comparison will help us assess how well LLMs perform in Arabic stance detection tasks.

The first step in this process will be Prompt Design, which is simply about determining how we will interact with the model to get the best results. Crafting clear and precise prompts is crucial because they directly influence the model's performance and accuracy in producing the desired output. For example, when asking the model to classify a tweet about the stance on women's sports in Saudi Arabia, we need to phrase the prompt carefully. An example could be: 'Is this tweet positive or negative about women's participation in sports? Do not explain or justify your answer. This is important because we are primarily interested in the label (positive, negative, or neutral) for classification purposes, not in the explanation. The prompt structure controls how the model interprets the task and provides the result.

Next, we will move on to experimenting with Zero-Shot and Multi-Shot. In Zero-Shot, we will ask the model to classify tweets directly without providing it with previous examples. The model will rely on its general knowledge for classification. For example, if a tweet says, 'مبروك لسيّدات النصر الفوز بالدوري وإبارك خصيصة لبناتنا السعوديات وفقكّن الله' the model will classify it as positive based on its understanding.

In multi-shot, we will provide the model with several examples of how to classify tweets. For instance, 'Tweet 1: رؤية المرأة السعودية في الاولمبياد يدعوا للفخر. Classification: Positive' and 'Tweet 2: حجاب البنّت هو اساس عفتها منظر لاعبات الكورة مقزز. Classification: Negative.' This method helps guide the model by giving clear examples of what constitutes a positive or negative stance. The idea behind multi-shot is that the more examples we provide, the better the model will understand the task, leading to more accurate classifications.

This phase will be crucial because it will allow us to use AI models effectively and efficiently to analyze public opinions on specific topics like women's sports. Instead of manual classification, which takes a lot of time, we will be able to use these models to process large amounts of tweets quickly, saving time and effort while providing more accurate and reliable results.

Chapter 4 Results and Discussion

4.1 Evaluation Metrics

In evaluating stance detection models, the focus is on measuring the model's ability to correctly classify stances into categories such as 'supportive,' 'opposed,' and 'neutral.' A range of metrics such as accuracy, precision, recall, and F1 Score for Its importance lies in the unbalanced data, which is a problem we face.

- **Accuracy** is calculated as the proportion of correct predictions:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative.

- **Precision** measures the proportion of true positive predictions out of all positive predictions:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- **Recall** calculates the proportion of true positive predictions out of all actual positives:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- **F1 Score** is the harmonic mean of precision and recall:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

To evaluate performance across multiple categories, we can also use the **macro average** for precision, recall, and F1 score. The macro average is calculated by taking the average of the metrics across each category without considering class imbalance:

$$\text{Macro Average} = \frac{1}{n} \sum_{i=1}^n \text{Metric}_i$$

4.2 Results

MODEL	MODEL USAGE STRATEGY	F1-MACRO	F1-POS	F1-NEG	F1-NAT
ARABERT (BASELINE)	Embeddings + Logistic Regression	0.9323	0.9435	0.9433	0.9100
ARABERT (FINE-TUNED)	Full Fine-tuning for classification	0.9228	0.9503	0.9315	0.8867
ALLAM (0-SHOT)	Constrained Decoding	0.3791	0.6209	0.5165	0.0000
ALLAM (1-SHOT)	Constrained Decoding	0.3232	0.6097	0.3599	0.0000
ALLAM (3-SHOT)	Constrained Decoding	0.3435	0.6154	0.3836	0.0316
LORA-ALLAM	LoRA + Sequence Classification	0.9125	0.9232	0.9254	0.8889
LLAMA-3.1 (0-SHOT)	Constrained Decoding	0.2527	0.5857	0.1046	0.0676
LLAMA-3.1 (1-SHOT)	Constrained Decoding	0.5465	0.6711	0.7262	0.2422
LLAMA-3.1 (3-SHOT)	Constrained Decoding	0.5800	0.7165	0.6966	0.3270

Table 3 Performance Comparison of Evaluated Models for Arabic Stance Detection

Table 3 presents a comprehensive comparison of the evaluated models for Arabic stance detection in discussions related to women’s sports. The results reveal clear performance differences between traditional supervised transformer-based classifiers, instruction-tuned large language models (LLMs) used in zero-, one-, and three-shot prompting settings, and a parameter-efficient fine-tuned LLM.

The AraBERT Baseline, which relies on frozen AraBERT embeddings combined with a Logistic Regression classifier, achieved the highest overall performance (F1-Macro = 0.9323). This indicates that contextual Arabic embeddings, even without additional fine-tuning, are highly effective for stance classification when used with a simple supervised classifier. This strong performance can be attributed to AraBERT’s pretraining on large-scale Arabic corpora, enabling it to capture rich linguistic and semantic representations well suited for this task.

The fully fine-tuned AraBERT model also achieved strong results (F1-Macro = 0.9228), though slightly lower than the baseline. This marginal underperformance may be attributed to the sensitivity and instability commonly observed when fine-tuning pre-trained language models on limited data. The relatively small dataset size (1348 tweets) may have limited the potential improvement from fine-tuning and introduced minor overfitting. This observation aligns with prior research highlighting that fine-tuning performance can be highly sensitive to hyperparameter settings and training conditions in low-resource scenarios. Zhang et al. [10] (2021) demonstrated that fine-tuning outcomes vary significantly depending on optimization settings and data size, which may prevent models from reaching their optimal performance when the dataset is small. Despite this, the model maintained balanced performance across all classes, including the minority Neutral class.

In contrast, instruction-tuned LLMs such as **ALLaM** and **LLaMA 3.1**, when used with constrained decoding in zero-, one-, and three-shot prompting, demonstrated significantly weaker performance. Notably, ALLaM struggled to recognize the Neutral class, frequently predicting only Positive or Negative labels. This behavior suggests that prompt-based classification without task-specific adaptation is insufficient for capturing nuanced stance in Arabic social media text.

Llama 3.1 showed relatively better performance in the few-shot setting, particularly in the three-shot configuration (F1-Macro = 0.5800), where a more balanced class distribution began to appear. However, its performance remained considerably lower than that of the supervised AraBERT models, indicating that general instruction-following capabilities do not directly translate into accurate stance classification in Arabic without fine-tuning.

A key finding of this study is the effectiveness of parameter-efficient fine-tuning. Applying LoRA to ALLaM for sequence classification (LoRA-ALLaM) significantly improved performance (F1-Macro = 0.9125), bringing the model close to AraBERT’s results. This demonstrates that the primary limitation of LLMs in zero-shot settings is not model capacity, but rather the absence of task-specific adaptation. Once adapted using LoRA, the LLM became capable of producing balanced and accurate predictions across all stance categories.

Overall, the results confirm that task-specific supervised training remains more reliable than prompt-based inference for Arabic stance detection. Furthermore, Arabic-specialized models such as AraBERT remain highly competitive even when compared with large LLMs. Most importantly, parameter-efficient fine-tuning techniques like LoRA can effectively unlock the true potential of LLMs for Arabic NLP tasks.

To further understand model behavior beyond numerical metrics, a confusion matrix is used to analyze how predictions are distributed across stance categories. This analysis highlights the types of misclassifications made by each model and identifies the categories where performance weaknesses are most apparent. Such insights are essential for conducting a detailed error analysis and guiding future model improvements. Here are some examples:

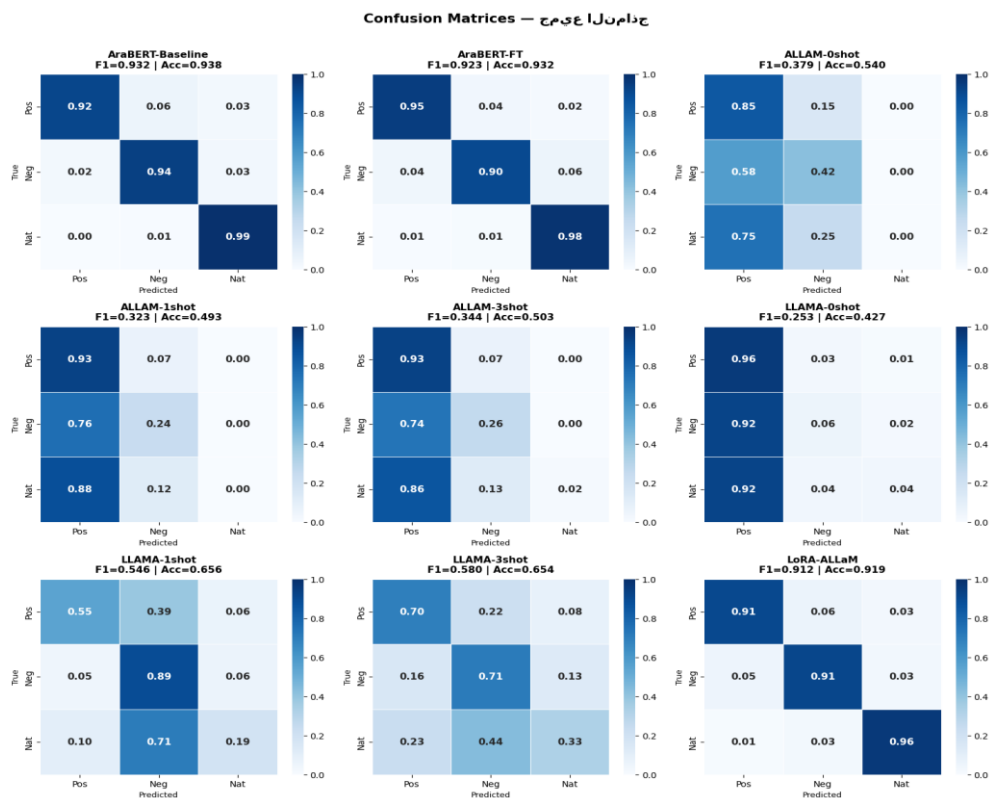


Figure 2 Confusion Matrices of AraBERT and ALLaM and LLaMA

MODEL	POS (P)	POS (R)	POS (F1)	NEG (P)	NEG (R)	NEG (F1)	NAT (P)	NAT (R)	NAT (F1)
ARABERT BASELINE	0.97	0.92	0.94	0.95	0.94	0.94	0.84	0.99	0.91
ARABERT FINE-TUNED	0.95	0.95	0.95	0.96	0.90	0.93	0.81	0.98	0.89
LORA-ALLAM	0.94	0.91	0.92	0.94	0.91	0.93	0.83	0.96	0.89
LLAMA (3-SHOT)	0.73	0.70	0.72	0.68	0.71	0.70	0.33	0.33	0.33
LLAMA (1-SHOT)	0.87	0.55	0.67	0.61	0.89	0.73	0.33	0.19	0.24
LLAMA (0-SHOT)	0.42	0.96	0.59	0.59	0.06	0.10	0.29	0.04	0.07
ALLAM (3-SHOT)	0.46	0.93	0.62	0.73	0.26	0.38	0.43	0.02	0.03
ALLAM (1-SHOT)	0.45	0.93	0.61	0.71	0.24	0.36	0.00	0.00	0.00
ALLAM (0-SHOT)	0.49	0.85	0.62	0.66	0.42	0.52	0.00	0.00	0.00

Table 4 Detailed Per-Class Performance of All Models Using Precision, Recall, and F1-Score

Table 4 presents a detailed per-class evaluation of the best-performing models using Precision, Recall, and F1-score for each stance category (Positive, Negative, and Neutral). Unlike overall metrics, this breakdown provides deeper insight into how each model handles class imbalance and reveals where specific weaknesses occur.

The results show that the AraBERT Baseline and AraBERT Fine-Tuned models achieved consistently high Precision and Recall across all classes. Both models demonstrated strong Recall for the Neutral class, though Precision was relatively lower, suggesting occasional misclassification of non-neutral instances as Neutral. This indicates that AraBERT’s contextual Arabic representations are highly effective in capturing subtle linguistic cues associated with neutral stance.

Similarly, the LoRa Allam model achieved balanced performance across the three classes, with especially strong Recall for the Neutral class. This confirms that parameter-efficient fine-tuning enabled the LLM to properly learn stance-specific patterns that were not captured during zero-shot or few-shot prompting. In contrast, the zero-shot and one-shot configurations of ALLaM and Llama 3.1 showed very poor Recall for the Neutral class, in some cases close to zero. This explains the low F1-Macro scores observed earlier and confirms that these models tended to ignore the minority class and over-predict the dominant classes (Positive or Negative). Even when Precision appeared reasonable for certain classes, the low Recall indicates that many true instances were missed. For instance, ALLaM (0-shot) recorded a Neutral Recall of 0.00, indicating a complete failure to detect any Neutral stance.

The three-shot configuration of Llama 3.1 showed noticeable improvement compared to zero- and one-shot settings, achieving more balanced Recall across classes. However, its performance remained significantly lower than that of the supervised and fine-tuned models. This per-class analysis reveals critical performance differences that would otherwise be

masked by overall metrics alone. For example, ALLaM (0-shot) achieved an accuracy of 0.5401, yet its Neutral F1-score was 0.00, demonstrating a complete failure to detect the minority class despite the seemingly moderate overall score.

4.3 Error analysis

This section analyses the errors made by the models to understand their limitations and identify challenging cases in stance detection. Several factors may have contributed to the relatively low performance of the instruction-tuned LLMs, particularly ALLaM and LLaMA in the zero-shot and one-shot and three-shot settings.

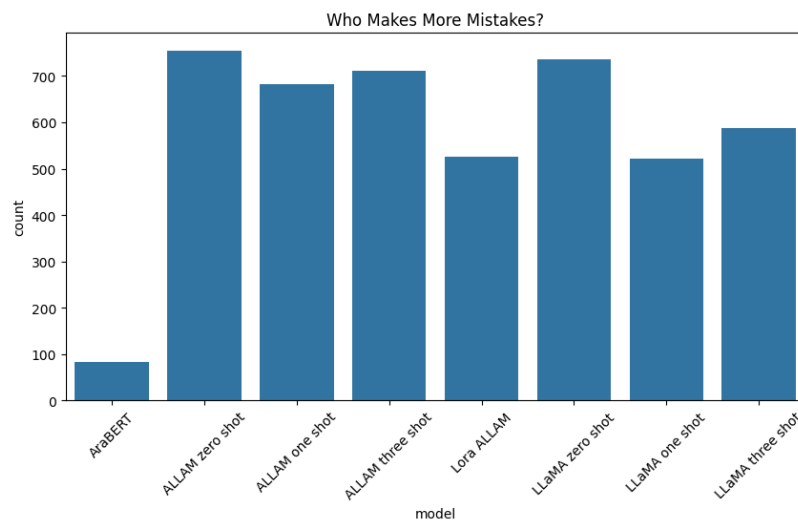


Figure 3 Distribution of Classification Errors Across Models

Figure3 shows the distribution of errors across models. The zero-shot versions of both ALLaM and LLaMA produce the highest number of misclassifications, indicating their limited ability to generalize without task-specific guidance.

In contrast, the BERT baseline achieves the lowest number of errors, followed by the fine-tuned ALLaM (LoRA). This suggests that task-specific fine-tuning significantly improves performance compared to prompt-based approaches, especially in structured classification tasks like stance detection.

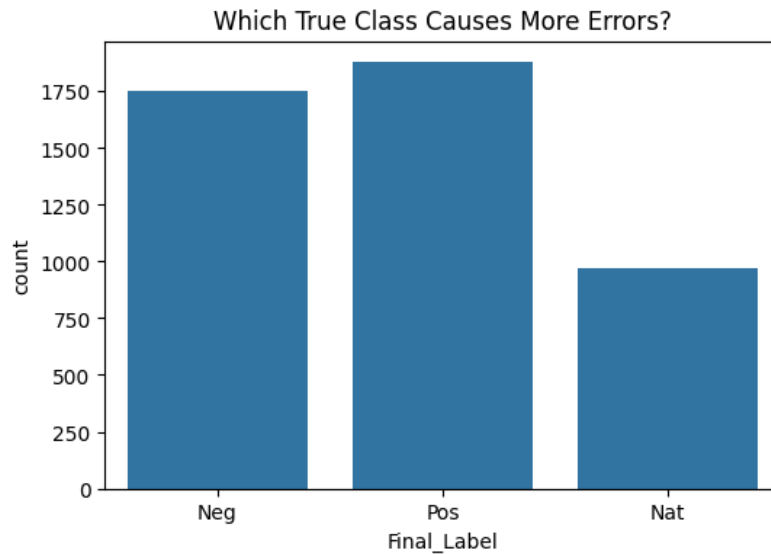


Figure 4 Classes Contributing Most to Misclassification

This plot shows the distribution of errors across classes. Although the positive (pos) class appears to contribute the highest number of errors, followed by the negative (neg) class, this observation is influenced by class imbalance in the dataset. The higher error count in the positive class does not necessarily indicate that it is inherently more difficult, but rather that it has more samples overall.

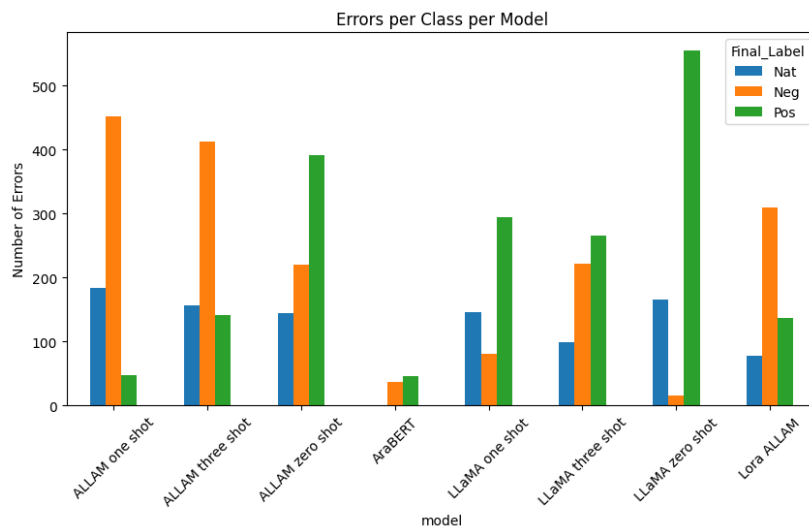


Figure 5 Error Analysis per Class Across Different Models

This plot presents the distribution of errors per class for each model. The results show that the fine-tuned ALLAM model (LoRA), as well as the shot-based settings, still struggle with negative samples, suggesting that detecting negative stance remains challenging for ALLAM even after adaptation.

In contrast, the LLaMA model exhibits a higher tendency to misclassify positive comments, reflecting difficulty in accurately identifying positive stance.

Meanwhile, the BERT baseline produces significantly fewer errors across all classes, demonstrating strong overall performance in this task.

These findings suggest that different models struggle with different aspects of stance detection. While ALLAM faces challenges in identifying negative sentiment, LLaMA appears to have difficulty distinguishing positive expressions, potentially due to variations in linguistic patterns or class distribution within the dataset.

The following examples highlight challenging cases where models fail to correctly identify stance, particularly in ambiguous or context-dependent situations.

4.3.1 AraBERT Errors

predicted	Final_Label	text
Nat	Pos	من وجهة نظري، اللاعنات السوديت: فيه عناس تيشّر لكن يفتاحون فيت وتدريب واجتهاد أكبر
Neg	Pos	طبيب محسني دوري النساء له عشر سنين؟ الطبيعي انه كل شي جديد بيتا له تهيئة وبيئة اما كنا تجيب بذاك فاجه من الدم ويتعامم محترفات والمجتمع متقلّبم شي صعب مره ايدا بالإساسيات رياضة المدارس اكاديميات تجهيز المدارس بملاعب واشتلة ورياضية واكيد بيغير الجيل ومو عيش هالشي محد يشوفه عيب
Neg	Pos	الحدث هـ فزينا بالدوري مرتين لكل لعبة ولكل رياضة فالونها وتكتيك معين مو بس الحق الكوره وصلى الله وسلم اذا العيب يعني اذا التزم بقلون معين عليه اللجه مثل ما تمسكين دوامك من ٨ إلى ٣ بقلون معين المرء السعودية المسلمة تلعب بمكان مغلق ومع لسانا
Neg	Pos	##### والله كنو ياكنا جابوا الكوايتة بالإراض
Neg	Pos	مسكين الجمع يكره التشنج الديني حماد الله الأمير محمد بن سلمان خلصنا من التشدد من يصدق المرء السعودية تفقد سياره وعيادته وتلعب رياضة ويحلق بالفتنزيون ودخلت السلك العسكري والديبلوماسية مرجحاً بكم في السعودية الجديده

Figure 6 Sample Misclassified Tweets by AraBERT

Although the AraBERT baseline achieves higher overall performance with fewer errors, this result should be interpreted with caution. The dataset is imbalanced, with the positive class being more dominant, which may bias the model toward predicting this class more frequently.

In addition, AraBERT is specifically designed for classification tasks, allowing it to effectively capture surface-level patterns and keywords, particularly in short and explicit comments. This explains its strong performance in simpler cases where stance is clearly expressed. But a closer examination of misclassified samples reveals that AraBERT lacks deeper contextual understanding. It frequently fails on longer and more complex comments, where the stance is less explicit and relies more on context. In many of these cases, the model incorrectly predicts a negative stance simply because certain negative keywords appear in the text, without properly considering the overall context or meaning of the comment.

These findings suggest that while AraBERT performs well in scenarios driven by lexical cues and straightforward patterns, it is limited in handling context-dependent and linguistically complex inputs.

4.3.2 ALLaM Errors

= ALLaM zero shot =====			
text	Final_Label	predicted	
والله خراب يسير بسرعه خطيره الله المستعان الدجال على الابواب سحفا للمجرمين الذين ينسبون في بلاد الحرمين وسحفا للرياضه	Neg	Pos	
يوم مميز في تاريخ الرياضة السعودية	Pos	Neg	
تطور ملحوظ في مستوى المرأة السعودية في مجال الرياضة لكن يحتاج اهتمام اكثر والاضيق	Nat	Neg	
اقول اكثرتم كلام بين الحرام حرام والحلال حلال ولي يبغي الحرام بكرا اما شايب معجز تكسرت عظامه ولا في القر مايقى له الا اثر الذنب	Nat	Neg	
ياقي المنتخب السعودي لسيارات 🏆💛	Pos	Nat	
== ALLaM three shot =====			
text	Final_Label	predicted	
نعم المرأة هني الى مثلنا سدى ونبيها تظهر للمام هذا وضع اغلب السعوديات عمارس رياستها بحدود لكن لانصف اعلامنا لاينظر سوا الله مخلقه وثقه ويسندوا للمام على انها مهال المرأة السعودية	Pos	Neg	
لاكتطوا صور ساء فكله لانها ذنب استخر الله العظيم	Neg	Nat	
يللاز	Neg	Pos	
🤔🤔🤔🤔..	Nat	Pos	
هذه حقيقة السعودية عري باسم الرياضة 🤔🤔	Neg	Pos	
= ALLaM one shot =====			
text	Final_Label	predicted	
لا احد يفارهم بالرجال نول حدهم المطبخ	Neg	Pos	
اذا سعودي وهو مصدق اللي فاعد تشوفه عيوني على بناتنا واخواتنا لا حول ولا قوة الا بالله العلي العظيم	Neg	Pos	
🤔🤔🤔🤔🤔	Nat	Pos	
وين الحشمة يا بنت حسي الله على من خرب نظام المملكة والله اذا اعتربت فيها فرة ما كنت اشوف بنت غير محتشمة الان اشوفين في مواقع التواصل ولا اسندق عيوني كيف صارت	Neg	Pos	
مليانا من هريذكم	Neg	Pos	

Figure 7 Sample Misclassified Tweets by ALLaM

A common pattern observed in the misclassified samples is that many comments with a true negative label are predicted as positive or neutral by the model. In several cases clearly negative comments were misinterpreted as positive, indicating that the model fails to correctly capture the underlying stance. Some neutral or ambiguous expressions are incorrectly classified as negative, suggesting inconsistency in the model’s decision boundaries and this indicates that the model struggles to accurately determine stance, particularly when it is implicit or context dependent.

Also, emojis do not appear to assist in understanding the intended meaning. For example, a comment containing only a laughing emoji was classified as neutral, whereas it conveys sarcasm and mockery toward Saudi women’s sports.

This highlights the model’s limitation in interpreting non-textual cues and recognizing sarcastic intent.

The failure of ALLaM across all prompting configurations to predict the Neutral class ($F1 = 0.00$) is consistent with the label bias phenomenon documented by Aiyappa et al. (2024) [8], who showed that zero-shot LLMs tend to collapse predictions onto dominant classes, effectively ignoring minority labels such as Neutral in three-class classification tasks.

4.3.3 LLaMA Errors

LLaMA zero shot =====			
	text	Final_Label	predicted
	ما شاء الله إيجاز رياضي تاريخي يثبت مكانة المرأة السعودية وإسهامها على الصبر عالميا كل التوفيق في الهاهي	Pos	Neg
	مشاركة اليوم الوطني حتي السهه مخظنه حيث أفل فيها تطور الرياضة بين المادني والحاضر ويمكن المرأة السعوديه في مجال الرياضة وإثباته الفرصه لنا في هذا المجال، في ظل هيلندا الرشيدة تهينس SA ♥ الهمه ويسمى السءاء والتمنية لهذه الأرض المباركة ... حفظ الله هيلندا وأرضها المسءاء	Pos	Neg
	للف ميزوك يا عالميات دوم يارب متصبرات والذهب ماثلق الا يكم	Pos	Neg
	السعودية اولي! عندنا تلك مبدعات في الرياضة مثل هتان السيف، المها باهليل، دانية عقيل، سارة عبدالجواد وسمية مشي، كل وحدة عندها إيجاز رفيع راس الوطن. بس إيه يحطون ساره مهتد واختها في إحدت رياضي وهي ما لها دخل بالرياضة أصلاً؟	Pos	Neg
	أسأل الله لها التوفيق والسداد والحمد لله ... المرأة السعودية حظيت وتحظى باهتمام ورعاية القادة الحكمة ومنحتها سبل التمكين بالعديد من القرارات والمبادرات التاريخية لشكل نظرة نوعية وجبر مسبوقة لتبرز . دورها في مختلف الميادين، فهي شريك فاعل في رفعة الوطن وتماته، في مختلف المجالات علميا واقتصاديا وسوسيا واجتماعيا ورياضيا	Pos	Neg
= LLaMA three shot =====			
	text	Final_Label	predicted
	تستغرب من حثيدات خديجه وعائشه وحفصه وام سلمه هذه الجراء على الله سبحانه الرياضة مطلوبه ولكن الحشمه مطلوبه اكثر	Neg	Nat
	ختيه أشتياء أهم للمرأة السعودية من البطولات والمنقسات ترا	Neg	Nat
	النادي الوحيد في العالم اللي حريمهم اطلق من رجالهم .. كفو ياكسر	Pos	Neg
	لي كره القدم حرام ولا عيب خيي ظلي مين الي حرم كره القدم عن البيت خيي هديه هوايي وعم لديها الله ما حرم كره القدم عن البيت فدا تنهم وتجي تصلحها عنا تمام	Pos	Neg
	بطلات 🇸🇦	Pos	Nat
LLaMA one shot =====			
	text	Final_Label	predicted
	والله الحرام انت الي تشوف حرام ان رينا وضع الكيه في مكه اليه يعني الكيه في بيتك يلطام انت؟ كل واحد نتيه على جيبه من انت لتحابس الناس؟ وتكلم على لسان الله وتحرم وتحلل كان الله الحق في هذا! راجع لسلك الله بيبك	Pos	Neg
	القدام افضل للتصر	Pos	Neg
	النادي الوحيد في العالم اللي حريمهم اطلق من رجالهم .. كفو ياكسر	Pos	Neg
	إنا لله وإنا إليه راجعون . اللهم لا تؤاخذنا بما فعل السفهاء منا	Neg	Nat
	أسس شقت فترين عن اللاعبات السعوديات، أول مرة أروي إن عندنا فريق بهالمستوى	Nat	Pos

Figure 8 Sample Misclassified Tweets by LLaMA 3.1

A similar pattern is observed for the LLaMA model, but with errors concentrated in the positive class. In the zero-shot setting, the model tends to make more errors on longer and more complex comments, indicating difficulty in handling extended context. While performance slightly improves in the one-shot setting, new error patterns emerge, where positive comments are sometimes misclassified as negative or neutral. This suggests that the model struggles to recognize supportive or approving stance when it is expressed implicitly.

Additionally, comments containing emojis that convey support are often labeled as neutral. This may be because the model is unable to link these expressions to the broader context or interprets them as vague and lacking sufficient information to assign a clear stance. As a result, such comments are treated as general or ambiguous rather than explicitly positive. Overall, although prompting provides some guidance, it does not fully resolve the model’s limitations in stance interpretation.

4.3.4 LoRA-ALLaM Errors

#	Lora ALLAM =====				
		text	Final_Label	predicted	
		كيف يجبروني؟؟ ما اهديت لأسفل الرياضة فليس اهم بمشاركة المرأة السعودية فيها؟؟ أن كان هذا كسده و ان كان كسدك الرياضة بشكل علم فليس هناك أمل حتى و هي رياضة نفتت جمالها حين أصبحت لديه و لم تحصل حتي على الاهتمام رغم ذلك	Neg	Nat	
		واضح افهم خكاريه لو مسموح يلعبون مع دوري السيدات يروح ويسلي من كثرة التيلار واليونكس	Neg	Nat	
		اجل رويحي تليمي دوري السيدات وخلى روشن لرجاله	Neg	Nat	
		كلو سيدناكتا ههههههههههههههههههههه	Pos	Nat	
		من وجهة نظري، اللاعبات السعوديات فيه عئاس كثير لكن يحتاجون ذلك وتدريب واجتهاد اكبر	Pos	Nat	

Figure 9 Sample Misclassified Tweets by LoRA-ALLaM

The analysis of misclassified samples shows a consistent pattern where both positive and negative comments are frequently predicted as neutral. This suggests that the model tends to default to the neutral class when the stance is not explicitly clear or when the input contains mixed or ambiguous opinions. Many of the misclassified instances involve comments that express multiple viewpoints simultaneously or contain implicit attitudes that are not directly stated. In these cases, the stance is not straightforward, but rather embedded within sarcasm, layered opinions, or socially contextualized discussions. As a result, the model struggles to determine a single dominant stance and assigns a neutral label.

In contrast, human interpretation can infer meaning beyond the literal text by considering context, tone, and underlying intent. For example, discussions related to women's sports often carry social, cultural, and emotional dimensions, where the stance may not be explicitly stated but is understood through context and shared knowledge.

4.4 Limitations

Several limitations affected the reliability and generalizability of the models used in this study. Arabic stance detection remains a highly challenging task due to the scarcity of high-quality dialectal datasets, the subjective nature of annotation, and the sensitivity of large language models to prompt design. In addition, social media comments often contain sarcasm, indirect language, and culturally dependent expressions that cannot always be interpreted through straightforward positive, negative, or neutral categories. The following sections discuss the main limitations encountered throughout this research and how they may have influenced the experimental results.

4.4.1 Data Scarcity and Arabic Language Representation

One of the primary limitations is the scarcity of high-quality annotated Arabic data, particularly for dialectal varieties. The dataset used in this study consists of Saudi dialect tweets, a variety that remains significantly underrepresented in the pretraining corpora of most large language models. This challenge is not unique to our work; Alkhatlan et al. (2025) similarly highlights that the lack of comprehensive Arabic datasets has limited research

progress in Arabic NLP, noting that most existing resources remain limited in terms of topic diversity and dialectal coverage. Furthermore, Alkhraji and Azmi (2025) explicitly selected traditional ML classifiers over deep learning approaches partly due to limited annotated Arabic data availability, which they identified as a barrier to effective deep learning training.

4.4.2 Annotation Complexity and Subjectivity

Stance detection in Arabic social media is inherently difficult to annotate consistently, as opinions are frequently expressed through indirect language, sarcasm, and culturally specific references. As noted in the Arabic stance detection literature, the diglossic reality of Arabic social media — where tweets blend Modern Standard Arabic with regional dialects — significantly complicates the annotation process, as a single word can carry contradictory meanings across different dialects. This ambiguity introduces noise into the dataset and makes it particularly challenging for models to learn consistent patterns, especially for borderline Neutral cases. The MAWQIF dataset annotation process, for instance, required multiple iterations and a panel of three to seven annotators per tweet to ensure reliability, reflecting the inherent difficulty of achieving consistent stance labels in Arabic text.

4.4.3 Prompt Design

Perhaps the most significant contributing factor to the underperformance of LLMs in this study is the quality and structure of the prompts used. Prompt design directly determines how well an LLM interprets and executes a classification task, and suboptimal prompting can substantially suppress model performance regardless of the model's underlying capacity. This sensitivity to prompt formulation has been documented in prior work on Arabic stance detection: Cruickshank and Ng (2024), as cited in the StanceEval 2024 overview, found that while few-shot LLM prompting results can be competitive with supervised models, these results often show inconsistent performance, raising questions about whether LLMs should be employed for stance detection at all. Moreover, the SMASH team at StanceEval 2024 experimented with different Arabic and English prompts and found that even the choice of language used in the prompt affected results, with English prompts outperforming Arabic ones — highlighting how sensitive LLM performance is to prompt formulation choices. In our experiments, ALLaM consistently failed to predict the Neutral class across all prompting configurations, suggesting that the prompts did not sufficiently convey what constitutes a neutral stance in this specific context. More carefully designed prompts that explicitly define the Neutral category with representative examples may have meaningfully improved performance.

Chapter 5 Conclusion and Future Work

In conclusion, women's sports in Saudi Arabia have become a genuinely contested social topic, and this project set out to understand that conversation through the lens of AI. The team built a stance detection system from the ground up collecting over 1,300 Arabic comments from X and YouTube, manually labeling each one as supportive, opposing, or neutral, and then testing a range of modern language models to see which could best understand what people were saying.

What made this work interesting and difficult is the nature of Arabic social media itself. People use dialect, sarcasm, emojis, and culturally loaded language in ways that don't translate neatly into positive or negative labels. The team kept emojis in the data on purpose, recognizing that a laughing emoji on a sensitive topic means something very different from its face value.

The results demonstrated clear differences between models. AraBERT, the Arabic-specialized model, handled the task remarkably well without needing much adaptation, reaching an F1 of 0.93. Large language models like ALLaM and LLaMA, when simply prompted without fine-tuning, struggled badly especially with neutral content, which some missed entirely. However, once ALLaM was fine-tuned using LoRA, its performance improved significantly, which suggests the gap wasn't really about intelligence it was about preparation. A model that hasn't been shown what the task looks like will underperform no matter how powerful it is.

In future work, expanding the dataset by including additional social media platforms, broader time periods, and more dialectal variation would improve model robustness. Another major challenge is sarcasm detection, which remains difficult for most models. Many sarcastic comments were misclassified because models interpreted them literally. Addressing this issue may require either specialized training data or dedicated sarcasm detection components.

Additionally, prompt engineering deserves more attention in future studies. The way prompts are designed significantly affects model performance, and further experiments could explore structured reasoning prompts or compare performance between Arabic and English prompting.

Beyond model improvements, there's a bigger opportunity here. Public opinion on women's sports in Saudi Arabia is changing in real time, tied directly to Vision 2030 milestones. A system capable of tracking those shifts over time could provide valuable insights for policymakers, sports organizations, and researchers trying to understand how society is responding to these changes. That's the direction worth building toward.

References

- [1] A. K. Alkhraiji and A. M. Azmi, “Stance detection in Arabic tweets: A machine learning framework for identifying extremist discourse,” *Mathematics*, vol. 13, no. 2965, 2025. [Online]. Available: <https://doi.org/10.3390/math13182965>
- [2] S. M. Mohammad, P. Sobhani, and S. Kiritchenko, “Stance and sentiment in tweets,” *ACM Transactions on Internet Technology*, vol. 17, no. 3, p. 26, 2016. [Online]. Available: <https://doi.org/10.1145/3003433>
- [3] K. Darwish, P. Stefanov, M. Aupetit, and P. Nakov, “Unsupervised user stance detection on Twitter,” in *Proc. Int. AAAI Conf. Web Social Media (ICWSM)*, vol. 15, no. 1, pp. 186–197, 2021.
- [4] A. Alkathlan, F. Alahmadi, F. Kateb, and H. Al-Khalifa, “Constructing and evaluating ArabicStanceX: A social media dataset for Arabic stance detection,” *Frontiers in Artificial Intelligence*, vol. 8, 1615800, 2025. [Online]. Available: <https://doi.org/10.3389/frai.2025.1615800>
- [5] A. Charfi, M. Bessghaier, A. Atalla, R. Akasheh, S. Al-Emadi, and W. Zaghouani, “Stance detection in Arabic with a multi-dialectal cross-domain stance corpus,” *Social Network Analysis and Mining*, vol. 14, no. 161, 2024. [Online]. Available: <https://doi.org/10.1007/s13278-024-01335-5>
- [6] A. Alturayef, S. Alqahtani, and H. Alsubaie, “StanceEval 2024: The first Arabic stance detection shared task,” in *Proc. 2nd Arabic Natural Language Processing Conf.*, pp. 792–798, 2024.
- [7] M. Ibrahim, “CUFE at StanceEval 2024: Arabic stance detection with fine-tuned Llama-3 model,” in *Proc. 2nd Arabic Natural Language Processing Conf.*, pp. 807–810, 2024.
- [8] Aiyappa, R., Senthilmani, S., An, J., Kwak, H., & Ahn, Y. (2024). *Benchmarking zero-shot stance detection with FlanT5-XXL: Insights from training data, prompting, and decoding strategies into its near-SoTA performance*. arXiv preprint arXiv:2403.00236.
- [9] A. Jaziriyani, A. Akbari, and H. Karbasi, “ArabicStanceX: A general target-based stance detection corpus in Arabic,” *Proc. Arabic Natural Language Processing Conf.*, pp. 1–12, 2022.
- [10] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “Revisiting Few-sample BERT Fine-tuning,” arXiv preprint arXiv:2006.05987, 2020.

Appendixes

Team Member	Responsibilities
Noura Alhuthali (Team Leader)	• Coordinated team tasks and managed workflow. • Structured and organized the full report. • Collected dataset from X (Twitter) and YouTube. • Conducted error analysis. • Reviewed and ensured consistency across all sections. • Participated in all sections.
Lamar Allihyani	• Performed data cleaning and preprocessing. • Led the manual annotation process and managed labeling workflow. • Prepared examples of prompt structures used for stance classification. • Participated in all sections.
Areej Alsaari	• Developed the modeling methodology section. • Prepared AraBERT baseline and LLM comparison framework. • Calculated inter-annotator agreement and prepared dataset statistics. • Designed Zero-shot and Few-shot prompt strategies. • Participated in all sections.
Shahad Alsaari	• Wrote the Evaluation chapter and metric formulas (Accuracy, Precision, Recall, F1, Macro-F1) • Finalized the Introduction, Conclusion, and Future Work chapters. • Prepared tables and figures for the report. • Participated in all sections.

GitHub Repository

The full implementation, source code, datasets, and additional project resources can be accessed through the GitHub repository provided below:

[GitHub Repository](#)